Data-Backed Journal-Style Replication of *Ambition Is Nothing Without Focus*

Replication Paper Project

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Abstract

This paper presents a local calculated replication of Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 489,193 observations across 200 entities, and the batch headline estimate is -0.0022 for Breadth x Org Recomb Exp log with p = 0.0076. The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: recombination breadth; organizational focus; inventive impact; patent citations; local reproduction

1 Introduction

This replication begins from a paper-first reading of Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks whether broad recombination becomes less productive when it leans on familiar routines,

unless organizational focus offsets that negative transfer. The study is worth revisiting because The replication matters because it targets a mechanism-heavy paper where the interaction structure, not only the sign of a main effect, is the core theoretical claim. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted. The executable evidence base used in this manuscript contains 489,193 observations across 200 entities and spans 1986 to 2016 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets. The paper's contribution is a near-structure-preserving reproduction that shows how the source mechanism survives in a fully local cleaned patent panel. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions. The paper belongs to a top-journal literature on exploration, recombination, and the organizational conditions under which broad search helps or hurts inventive performance. [5, 2, 1, 4, 6]. That literature matters because the replication is not asking whether breadth is always good; it is asking whether the interaction between breadth, organizational experience, and focus survives in a fully executable patent panel. [5, 2, 1, 4, 6]. The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, and performance, and it evaluates whether that stream's central mechanism remains persuasive after the current local dataset translation is made explicit. [3, 5, 2, 1, 4, 6].

2 Original Study, Theory, and Replication Boundary

The focal source article is [3]. The original study can be summarized as a mechanism paper with a sharp empirical core. The source paper studies the interaction of breadth, organizational experience, and focus. Forward citations proxy inventive impact. The unit of analysis is the patent. That combination matters because the source article asks readers to believe not merely that two

variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data. In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue. The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design.

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The interaction structure is the theoretical core. The local reconstruction therefore adopts the following bounded decision: The local rerun keeps the same interaction logic on the clean GVKEY-based panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Forward citations measure impact. The local reconstruction therefore adopts the following bounded decision: The local impact measure is the executable citation outcome already stored in the panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The paper works at the patent level. The local reconstruction therefore adopts the following bounded decision: The sample is narrowed to active S&P 500 utility

patents with unambiguous assignment. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline.

DirectSourceFact	LocalReplicationDecision
The interaction structure is the theoretical core.	The local rerun keeps the same interaction logic on the clean GVKEY-based panel.
Forward citations measure impact.	The local impact measure is the executable citation outcome already stored in the panel.
The paper works at the patent level.	The sample is narrowed to active S&P 500 utility patents with unambiguous assignment.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Clean GVKEY-anchored S&P 500 utility-patent panel from the local replication pipeline. PatentView-derived application year and citation outcomes. CRSP-Compustat linked firm-year filters and sector information already present in the workspace. Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 489,193 observations across 200 entities and covers 1986 to 2016 in the local executable panel. Every numeric claim in the results

section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article. This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated. The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper. The local sample frame is Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline The observed time window in the executable panel is 1986 to 2016 in the local executable panel The outcome is Patent-level inventive impact measured from forward citations . The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function.

3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Ghosh, Martin, Pennings, and Wezel (2014). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument. The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels. The coefficient export used in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels. The sample summary used in this paper serves a specific empirical role: it is needed to audit sample construction and raw

grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels. Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim.

Stage	NeededFor	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	Yes	Executable local result file

Verified evidence layers used to support the paper's calculations.

3.2 Sample Reconstruction

The executable sample is constructed in three observable stages. The raw regression dataset begins with 499,486 patent-level observations. After the active-firm and patent-type merge, the panel still contains 499,486 rows, but 3,678 patents remain ambiguously assigned and therefore cannot be used for a clean firm-level reproduction. The final analysis sample retains 489,193 observations after restricting attention to active S&P 500 utility patents with unambiguous GVKEY assignment.

This sample-construction sequence is important for interpretation. The final panel is not simply a large patent file; it is the outcome of a specific replication decision to privilege clean firm assignment over broader but noisier coverage. That decision preserves the organizational mechanism studied in the source paper, even though it may narrow the empirical frame relative to the original archive.

Step	Rows	UniqueGVKEY	UniqueYears
regression_dataset_raw	499,486	204	31
after_active_and_patent_type_merge	499,486	204	31
sp500_utility_drop_ambiguous	489,193	200	31

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Inventive impact is operationalized locally as Forward citations and appears in the executable dataset as inventive_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local cleaned citation outcome already stored in the regression dataset.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Breadth is operationalized locally as Centered breadth and appears in the executable dataset as breadth_c. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the centered breadth variable from the local regression output.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the speci-

fication instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Organizational experience is operationalized locally as Centered log prior organizational experience and appears in the executable dataset as org_recomb_exp_log_c. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the centered log experience variable from the same panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Organizational focus is operationalized locally as Centered firm-year focus and appears in the executable dataset as org_focus_c. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the contemporaneous focus measure already reconstructed in the local pipeline.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Controls is operationalized locally as Prior-art citations, degree, field experience, scale, age and appears in the executable dataset as n_prior_art_cites; recomb_degree; field_recomb_exp_log; total_patents_same_year; firm_age. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the local patent-level controls and fixed effects.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability

have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity.

Construct	Local measure	Derivation
Inventive impact	Forward citations	Use the local cleaned citation outcome already stored in the regression dataset.
Breadth	Centered breadth	Carry forward the centered breadth variable from the local regression output.
Organizational experience	Centered log prior organizational experience	Carry forward the centered log experience variable from the same panel.
Organizational focus	Centered firm-year focus	Use the contemporaneous focus measure already reconstructed in the local pipeline.
Controls	Prior-art citations, degree, field experience, scale, age	Reuse the local patent-level controls and fixed effects.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Ghosh, Martin, Pennings, and Wezel (2014). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Inventive impact, the manuscript uses Forward citations under the exact local implementation inventive_impact. The derivation step is: Use the local cleaned citation outcome already stored in the regression dataset.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Breadth, the manuscript uses Centered breadth under the exact local implementation breadth_c. The derivation step is: Carry forward the centered breadth variable from the local regression output.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Organizational experience, the

manuscript uses Centered log prior organizational experience under the exact local implementation org_recomb_exp_log_c. The derivation step is: Carry forward the centered log experience variable from the same panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Organizational focus, the manuscript uses Centered firm-year focus under the exact local implementation org_focus_c. The derivation step is: Use the contemporaneous focus measure already reconstructed in the local pipeline.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Controls, the manuscript uses Prior-art citations, degree, field experience, scale, age under the exact local implementation n_prior_art_cites; recomb_degree; field_recomb_exp_log; total_patents_same_year; firm_age. The derivation step is: Reuse the local patent-level controls and fixed effects.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Inventive impact has mean 6.2253, standard deviation 22.9068, and an interquartile range from 1.0000 to 6.0000; Breadth has mean -0.0064, standard deviation 5.2756, and an interquartile range from -3.0495 to 0.9505; Organizational experience has mean 0.0070, standard deviation 1.3465, and an interquartile range from -0.7782 to 0.3204; Organizational focus has mean -0.0004, standard deviation 0.0301, and an interquartile range from -0.0088 to -0.0017 . The time-series view is equally informative. Patents changes from 7306.0000 in the first observed period to 539.0000 in the last observed period; MeanBreadth changes from -1.9726 in the first observed period to -0.4169 in the last observed period; MeanOrgExperience changes from -0.7782 in the first observed period to 1.3178 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output. The regressor-quantile analysis shows that mean outcomes range from 4.9416 to 10.3793 across the support of the focal regressor, while the mean regressor itself ranges from -2.5926 to 6.7691. That

pattern is useful because it shows what the formal model is trying to account for in the current dataset. The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is.

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. Inventive impact has missing share 0.00%; Breadth has missing share 0.00%; Organizational experience has missing share 0.00% . At the other end of the support, Inventive impact remains close to complete with missing share 0.00%; Breadth remains close to complete with missing share 0.00%; Organizational experience remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively. Coverage across entities is also visible in the current panel rather than assumed. INTERNATIONAL BUSINESS MACHS COR contributes 83,091 observations; GENERAL ELECTRIC CO contributes 28,950 observations; MICROSOFT CORP contributes 26,529 observations; HEWLETT PACKARD CO contributes 24,186 observations; MOTOROLA INC contributes 15,304 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities. The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is -2.5926 and the mean outcome is 4.9416 across 229,109 observations. In the highest observed quantile, the corresponding values are 6.7691, 10.3793, and 94,587 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells. These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because it targets a mechanism-heavy paper where the interaction structure, not only the sign of a main effect, is the core theoretical claim. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication.

Variable	Missing	MissingShare
Inventive impact	0	0.00%
Breadth	0	0.00%
Organizational experience	0	0.00%
Organizational focus	0	0.00%
Prior-art citations	0	0.00%
Recombination degree	0	0.00%
Field experience	0	0.00%
Same-year patents	0	0.00%
Firm age	0	0.00%

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	489,193	6.2253	22.907	1.0000	3.0000	6.0000	0	1190
Breadth	489,193	-0.0064	5.2756	-3.0495	-1.0495	0.9505	-3	130
Organizational experience	489,193	0.0070	1.3465	-0.7782	-0.7782	0.3204	-1	7
Organizational focus	489,193	-0.0004	0.0301	-0.0088	-0.0066	-0.0017	0	1
Prior-art citations	489,193	18.918	57.508	5.0000	9.0000	16.000	2	2378
Recombination degree	489,193	0.5853	0.1778	0.4600	0.6122	0.7222	0	1
Field experience	489,193	1.3943	1.8625	0.0000	0.6931	2.4849	0	8
Same-year patents	489,193	1,106.6	1,468.4	228.00	495.00	1,281.0	1	7321
Firm age	489,193	16.322	7.3797	11.000	17.000	22.000	1	31

Main descriptive statistics discussed in the body of the paper.

Period	Patents	MeanBreadth	MeanOrgExperience	MeanImpact	Top5Share
1986	7,306	-1.9726	-0.7782	4.3848	5.49%
1987	7,920	-1.8485	-0.5598	4.4559	5.61%
1988	8,927	-1.7362	-0.4405	4.4998	5.43%

Period	Patents	MeanBreadth	MeanOrgExperience	MeanImpact	Top5Share
1989	9,876	-1.7341	-0.3318	4.7117	5.04%
1990	10,409	-1.7150	-0.2404	5.1830	5.24%
1991	10,950	-1.6663	-0.1752	5.6408	5.22%
1992	11,688	-1.5099	-0.1298	6.1589	5.43%
1993	12,054	-1.3694	-0.1058	6.4200	5.00%
1994	13,712	-1.2335	-0.0605	7.0965	5.08%
1995	16,332	-0.8868	0.0240	7.2316	5.00%
1996	15,923	-0.9553	0.0207	7.6327	4.97%
1997	18,799	-0.8085	0.0155	7.7720	5.36%
1998	18,164	-0.6502	-0.0664	7.6026	5.06%
1999	19,543	-0.3911	-0.0794	7.2368	5.22%
2000	22,175	-0.0682	-0.1024	6.7181	5.12%
2001	22,695	0.0757	-0.1184	5.4978	5.25%
2002	22,461	0.2464	-0.0957	5.3638	4.85%
2003	22,831	0.4267	-0.0539	5.1639	4.87%
2004	21,790	0.7716	-0.0321	5.3452	4.92%
2005	21,793	0.7341	-0.0008	5.2593	5.14%
2006	20,534	0.9402	0.0381	5.5693	4.91%
2007	22,148	1.1377	0.0755	5.7011	5.27%
2008	23,454	0.7233	0.1670	5.0370	5.40%
2009	17,971	1.1788	0.1283	5.1727	5.25%
2010	18,599	1.1069	0.1798	7.3907	5.01%
2011	19,134	0.9084	0.1868	8.3084	4.98%
2012	21,603	0.8664	0.3699	7.4903	5.04%
2013	16,415	0.7798	0.4202	7.2127	4.90%
2014	9,657	0.7923	0.5188	7.0003	4.90%
2015	3,791	0.5775	0.8984	5.9330	5.22%
2016	539	-0.4169	1.3178	3.7625	4.82%

Observed time patterns in the current dataset.

Quantile	N	MeanRegressor	MeanOutcome
Q1	229,109	-2.5926	4.9416
Q2	76,899	-1.0495	5.3861
Q3	88,598	0.3530	5.8387
Q4	94,587	6.7691	10.379

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Inventive impact	Breadth	Organizational experience	Organizational focus	Prior-art citations	Recombination degree	Field experience	Same-year patents	Firm age
Inventive impact	1.000	0.297	-0.016	-0.011	0.502	0.014	-0.027	0	0.005
Breadth	0.297	1.000	-0.209	-0.038	0.817	0.470	-0.303	0	0.167
Organizational experience	-0.016	-0.209	1.000	-0.025	-0.058	-0.383	0.867	0	0.155
Organizational focus	-0.011	-0.038	-0.025	1.000	-0.016	-0.057	0.008	0	-0.159
Prior-art citations	0.502	0.817	-0.058	-0.016	1.000	0.140	-0.107	0	0.114
Recombination degree	0.014	0.470	-0.383	-0.057	0.140	1.000	-0.492	0	0.120

Compact correlation structure for the main variables.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A patent-level count model with interactions among breadth, organizational recombination experience, and organizational focus, plus firm and year controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism. Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest. The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Active S&P 500 utility patents with unambiguous

GVKEY assignment in the local cleaned reproduction pipeline, and the empirical window is 1986 to 2016 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample. The second set of decisions concerns measurement and specification. The outcome is Patent-level inventive impact measured from forward citations, and the main explanatory design is Patent-level count model with breadth, organizational experience, organizational focus, and their interactions. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question. The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Prior-art citations, recombination degree, field experience, scale, age, plus firm and application-year fixed effects; clustered inference when available.. The crucial exclusion or substitution rule is: The executable sample drops ambiguous firm-patent assignments and limits the panel to locally verified active S&P 500 utility patents.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability. A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether broad recombination becomes less productive when it leans on familiar routines, unless organizational focus offsets that negative transfer.
Unit of analysis	Patent
Sample frame	Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline.
Time window	1986 to 2016 in the local executable panel.

DecisionPoint	LocalSpecification
Outcome definition	Patent-level inventive impact measured from forward citations.
Main explanatory design	Patent-level count model with breadth, organizational experience, organizational focus, and their interactions.
Controls and fixed effects	Prior-art citations, recombination degree, field experience, scale, age, plus firm and application-year fixed effects;
allowbreak clustered inference when available.	
Exclusion or substitution logic	The executable sample drops ambiguous firm-patent assignments and limits the panel to locally verified active S&P 500 utility patents.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline over 1986 to 2016 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Patent-level inventive impact measured from forward citations and the main explanatory design is Patent-level count

model with breadth, organizational experience, organizational focus, and their interactions. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Prior-art citations, recombination degree, field experience, scale, age, plus firm and application-year fixed effects; clustered inference when available. The main exclusion or substitution rule is The executable sample drops ambiguous firm-patent assignments and limits the panel to locally verified active S&P 500 utility patents. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$E[\text{Impact}_{pft} \mid Z_{pft}] = \exp\big(\beta_1 \text{Breadth}_{pft} + \beta_2 \text{OrgExp}_{pft} + \beta_3 \text{Focus}_{ft} + \beta_4 \text{Breadth}_{pft} \times \text{OrgExp}_{pft} + \beta_5 \text{OrgExp}_{pft} \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is -0.0022 on Breadth x Org Recomb Exp log (p = 0.0076). The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus. Breadth x org. experience is estimated at -0.0022 with p = 0.0076. Breadth x org. experience x focus is estimated at -0.1705 with p = 0.0152. Organizational experience is estimated at 0.0706 with p = 0.0461. The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Model	Term	Estimate	Std. error	P-value
Poisson Baseline	Breadth	0.0030	0.0158	0.8477
Poisson Baseline	Organizational experience	0.0706	0.0354	0.0461
Poisson Baseline	Organizational focus	-0.3609	0.3761	0.3373
Poisson Baseline	Breadth x org. experience	-0.0022	0.0008	0.0076

Model	Term	Estimate	Std. error	P-value
Poisson Baseline	Org. experience x focus	0.3219	0.1779	0.0703
Poisson Baseline	Breadth x org. experience x focus	-0.1705	0.0702	0.0152

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Breadth x org. experience is estimated at -0.0022 with a p-value of 0.0076; Breadth x org. experience x focus is estimated at -0.1705 with a p-value of 0.0152; Organizational experience is estimated at 0.0706 with a p-value of 0.0461; Org. experience x focus is estimated at 0.3219 with a p-value of 0.0703 . Within the focal specification, the sharpest local signal comes from Breadth x org. experience, which is estimated at -0.0022 with p = 0.0076. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment. The scale of the underlying variables also matters for interpretation. The outcome Inventive impact has mean 6.2253 and standard deviation 22.9068 in the local panel, while the main regressor Breadth has mean -0.0064 and standard deviation 5.2756. A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.9416 to 10.3793 while the mean regressor ranges from -2.5926 to 6.7691. This does not substitute for the model, but it shows how much raw variation the specification must explain. Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation.

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification. Within logit_top1, the leading coefficient family reads as follows: field_recomb_exp_log = -0.2116 (p = 0.0018); total_patents_same_year = -0.0002 (p = 0.0070); total_patents_until_prev_log = -0.5362 (p = 0.0097); Breadth x org. experience = -0.0132 (p = 0.0184); n_prior_art_cites = 0.0048 (p = 0.0274). The largest positive estimate in this specification is Organizational experience at 0.1380, while the largest negative estimate is Organizational focus at -0.9087. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure. Within logit_top5, the leading coefficient family reads as follows: Breadth x org. experience = -0.0118 (p = ¡0.001); total_patents_same_year = -0.0001 (p = ¡0.001); field_recomb_exp_log = -0.1319 (p = ¡0.001); total_patents_until_prev_log = -0.3860 (p = ¡0.001); recomb_degree = -0.3229 (p = 0.0168). The largest positive estimate in this specification is Organizational experience at 0.0988, while the largest negative estimate is Organizational focus at -0.8737. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure. Within poisson_baseline, the leading coefficient family reads as follows: Breadth x org. experience = -0.0022 (p = 0.0076); field_recomb_exp_log = -0.1139 (p = 0.0127); Breadth x org. experience x focus = -0.1705 (p = 0.0152); recomb_degree = -0.4546 (p = 0.0187); n_prior_art_cites = 0.0018 (p = 0.0268). The largest positive estimate in this specification is Org. experience x focus at 0.3219, while the largest negative estimate is recomb_degree at -0.4546. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure. This fuller coefficient reading is especially important for Ghosh, Martin, Pennings, and Wezel (2014) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text.

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Poisson Baseline, the most informative focal term is Breadth x org. experience with estimate -0.0022 and p = 0.0076; In Logit Top5, the most informative focal term is Breadth x org. experience with estimate -0.0118 and p = ¡0.001; In Logit Top1, the most informative focal term is Breadth x org. experience with estimate -0.0132 and p = 0.0184 . The Poisson Baseline specification contributes its own reading to the replication. Breadth x org. experience is estimated at -0.0022 with p = 0.0076 and is classified as a strong signal; Breadth x org. experience x focus is estimated at -0.1705 with p = 0.0152 and is classified as a moderate signal; Organizational experience is estimated at 0.0706 with p = 0.0461 and is classified as a moderate signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. The Logit Top5 specification contributes its own reading to the replication. Breadth x org. experience is estimated at -0.0118 with p = ¡0.001 and is classified as a strong signal; Organizational experience is estimated at 0.0988 with p = 0.0256 and is classified as a moderate signal; Breadth is estimated at 0.0147 with p = 0.3159 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. The Logit Top1 specification contributes its own reading to the replication. Breadth x org. experience is estimated at -0.0132 with p = 0.0184 and is classified as a moderate signal; Organizational experience is estimated at 0.1380 with p = 0.1395 and is classified as a weak signal; Breadth x org. experience x focus is estimated at -0.6392 with p = 0.5361 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. This model comparison matters for Ghosh, Martin, Pennings, and Wezel (2014) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition.

Term	Model	Estimate	PValue	Signal	Reading
Breadth	Poisson Baseline	0.0030	0.8477	Weak	Positive estimate with p = 0.8477.

Term	Model	Estimate	PValue	Signal	Reading
Organizational experience	Poisson Baseline	0.0706	0.0461	Moderate	Positive estimate with p = 0.0461.
Organizational focus	Poisson Baseline	-0.3609	0.3373	Weak	Negative estimate with p = 0.3373.
Breadth x org. experience	Poisson Baseline	-0.0022	0.0076	Strong	Negative estimate with p = 0.0076.
Org. experience x focus	Poisson Baseline	0.3219	0.0703	Suggestive	Positive estimate with p = 0.0703.
Breadth x org. experience x focus	Poisson Baseline	-0.1705	0.0152	Moderate	Negative estimate with p = 0.0152.
Breadth	Logit Top5	0.0147	0.3159	Weak	Positive estimate with p = 0.3159.
Organizational experience	Logit Top5	0.0988	0.0256	Moderate	Positive estimate with p = 0.0256.
Organizational focus	Logit Top5	-0.8737	0.5200	Weak	Negative estimate with p = 0.5200.
Breadth x org. experience	Logit Top5	-0.0118	¡0.001	Strong	Negative estimate with p = ¡0.001.
Org. experience x focus	Logit Top5	0.0361	0.9680	Weak	Positive estimate with p = 0.9680.
Breadth x org. experience x focus	Logit Top5	-0.2765	0.3557	Weak	Negative estimate with p = 0.3557.

Term	Model	Estimate	PValue	Signal	Reading
Breadth	Logit Top1	-0.0099	0.6686	Weak	Negative estimate with p = 0.6686.
Organizational experience	Logit Top1	0.1380	0.1395	Weak	Positive estimate with p = 0.1395.
Organizational focus	Logit Top1	-0.9087	0.8056	Weak	Negative estimate with p = 0.8056.
Breadth x org. experience	Logit Top1	-0.0132	0.0184	Moderate	Negative estimate with p = 0.0184.
Org. experience x focus	Logit Top1	-0.5147	0.8763	Weak	Negative estimate with p = 0.8763.
Breadth x org. experience x focus	Logit Top1	-0.6392	0.5361	Weak	Negative estimate with p = 0.5361.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

A separate model-fit export is not available for this paper, so the interpretation relies on the coefficient file and the observed distribution of the current dataset.

Model	Term	Estimate	Std. error	P-value
Poisson Baseline	Breadth	0.0030	0.0158	0.8477
Poisson Baseline	Organizational experience	0.0706	0.0354	0.0461
Poisson Baseline	Organizational focus	-0.3609	0.3761	0.3373
Poisson Baseline	n Prior Art Cites	0.0018	0.0008	0.0268
Poisson Baseline	Recomb Degree	-0.4546	0.1934	0.0187

Model	Term	Estimate	Std. error	P-value
Poisson Baseline	Field Recomb Exp log	-0.1139	0.0457	0.0127
Poisson Baseline	Total Patents Same Year	-0.0001	0.0001	0.1560
Poisson Baseline	Total Patents until Prev log	-0.1661	0.0839	0.0477
Poisson Baseline	Breadth x org. experience	-0.0022	0.0008	0.0076
Poisson Baseline	Breadth C:org Focus c	-0.0050	0.1059	0.9622
Poisson Baseline	Org. experience x focus	0.3219	0.1779	0.0703
Poisson Baseline	Breadth x org. experience x focus	-0.1705	0.0702	0.0152
Logit Top5	Breadth	0.0147	0.0147	0.3159
Logit Top5	Organizational experience	0.0988	0.0442	0.0256
Logit Top5	Organizational focus	-0.8737	1.3579	0.5200
Logit Top5	n Prior Art Cites	0.0036	0.0019	0.0556
Logit Top5	Recomb Degree	-0.3229	0.1351	0.0168
Logit Top5	Field Recomb Exp log	-0.1319	0.0343	0.0001
Logit Top5	Total Patents Same Year	-0.0001	0.0000	0.0000
Logit Top5	Total Patents until Prev log	-0.3860	0.1008	0.0001
Logit Top5	Breadth x org. experience	-0.0118	0.0023	0.0000
Logit Top5	Breadth C:org Focus c	-0.1306	0.3120	0.6754
Logit Top5	Org. experience x focus	0.0361	0.8998	0.9680
Logit Top5	Breadth x org. experience x focus	-0.2765	0.2994	0.3557
Logit Top1	Breadth	-0.0099	0.0231	0.6686

Model	Term	Estimate	Std. error	P-value
Logit Top1	Organizational experience	0.1380	0.0934	0.1395
Logit Top1	Organizational focus	-0.9087	3.6923	0.8056
Logit Top1	n Prior Art Cites	0.0048	0.0022	0.0274
Logit Top1	Recomb Degree	-0.4480	0.2543	0.0781
Logit Top1	Field Recomb Exp log	-0.2116	0.0677	0.0018
Logit Top1	Total Patents Same Year	-0.0002	0.0001	0.0070
Logit Top1	Total Patents until Prev log	-0.5362	0.2074	0.0097
Logit Top1	Breadth x org. experience	-0.0132	0.0056	0.0184
Logit Top1	Breadth C:org Focus c	-0.1517	1.2533	0.9036
Logit Top1	Org. experience x focus	-0.5147	3.3074	0.8763
Logit Top1	Breadth x org. experience x focus	-0.6392	1.0331	0.5361

Detailed coefficient table from the current local estimation output.

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. [3]. For Ghosh, Martin, Pennings, and Wezel (2014), the local headline result is Breadth x Org Recomb Exp log = -0.0022 with p = 0.0076. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than

asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper reports a negative interaction between recombination breadth and organizational experience, together with a focus-related moderation of that negative transfer.	Breadth x Org Recomb Exp log = -0.0022 with p = 0.0076.	This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel.
Measurement and data construction	The published analysis uses patent-level inventive impact and interaction terms among breadth, organizational experience, and organizational focus.	The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.
Sample and unit of analysis	The published design is a patent-level study of firm R&D using an archival panel of assigned patents and firm controls.	Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline. Local sample = 489,193 observations across 200 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published analysis uses patent-level inventive impact and interaction terms among breadth, organizational experience, and organizational focus By contrast, the local paper relies on the following executable design: The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace. The same caution applies to the sample itself. The published design is described as follows: The published design is a patent-level study of firm R&D using an archival panel of assigned patents and firm controls The local sample instead works from Active S&P 500 utility patents with unambiguous GVKEY assignment in the local cleaned reproduction pipeline and yields 489,193 observations across 200 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate. For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design. The comparison to the published paper is intentionally narrow. This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel. The manuscript therefore evaluates concordance

at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication. This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation.

8 Discussion

This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel. Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly. The first limitation concerns measurement. The local clean sample is paper-comparable rather than guaranteed identical to the archival source frame. The second limitation concerns sample translation. Patent-firm matching and sample filters follow the executable local pipeline. The third limitation concerns interpretation. Interpretation should emphasize mechanism recovery more than exact coefficient duplication.

9 Conclusion

The paper's contribution is a near-structure-preserving reproduction that shows how the source mechanism survives in a fully local cleaned patent panel. The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available.

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Step	Rows	UniquePatents	UniqueGVKEY	UniqueYears	AmbiguousPatents	Notes
regression_dataset_raw	499,486	495,460	204	31	NA	Patent-level regression dataset produced by Regression.R
after_active_and_patent_type_merge	499,486	495,460	204	31	3,678	Active panel duplicates on gvkey-year before tie-break: 0
sp500_utility_drop_ambiguous	489,193	489,193	200	31	0	Final clean sample used for reproduction

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	489,193	6.2253	22.907	1.0000	3.0000	6.0000	0	1190
Breadth	489,193	-0.0064	5.2756	-3.0495	-1.0495	0.9505	-3	130
Organizational experience	489,193	0.0070	1.3465	-0.7782	-0.7782	0.3204	-1	7
Organizational focus	489,193	-0.0004	0.0301	-0.0088	-0.0066	-0.0017	0	1
Prior-art citations	489,193	18.918	57.508	5.0000	9.0000	16.000	2	2378
Recombination degree	489,193	0.5853	0.1778	0.4600	0.6122	0.7222	0	1
Field experience	489,193	1.3943	1.8625	0.0000	0.6931	2.4849	0	8
Same-year patents	489,193	1,106.6	1,468.4	228.00	495.00	1,281.0	1	7321
Firm age	489,193	16.322	7.3797	11.000	17.000	22.000	1	31

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Inventive impact	Breadth	Organizational experience	Organizational focus	Prior-art citations	Recombination degree	Field experience	Same-year patents	Firm age
Inventive impact	1.000	0.297	-0.016	-0.011	0.502	0.014	-0.027	0	0.005
Breadth	0.297	1.000	-0.209	-0.038	0.817	0.470	-0.303	0	0.167
Organizational experience	-0.016	-0.209	1.000	-0.025	-0.058	-0.383	0.867	0	0.155
Organizational focus	-0.011	-0.038	-0.025	1.000	-0.016	-0.057	0.008	0	-0.159
Prior-art citations	0.502	0.817	-0.058	-0.016	1.000	0.140	-0.107	0	0.114
Recombination degree	0.014	0.470	-0.383	-0.057	0.140	1.000	-0.492	0	0.120
Field experience	-0.027	-0.303	0.867	0.008	-0.107	-0.492	1.000	0	0.094
Same-year patents	-0.018	-0.024	0.212	-0.182	-0.040	0.070	0.112	1	0.300
Firm age	0.005	0.167	0.155	-0.159	0.114	0.120	0.094	0	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Patents	MeanBreadth	MeanOrgExperience	MeanImpact	Top5Share
1986	7,306	-1.9726	-0.7782	4.3848	5.49%
1987	7,920	-1.8485	-0.5598	4.4559	5.61%
1988	8,927	-1.7362	-0.4405	4.4998	5.43%
1989	9,876	-1.7341	-0.3318	4.7117	5.04%
1990	10,409	-1.7150	-0.2404	5.1830	5.24%
1991	10,950	-1.6663	-0.1752	5.6408	5.22%
1992	11,688	-1.5099	-0.1298	6.1589	5.43%
1993	12,054	-1.3694	-0.1058	6.4200	5.00%
1994	13,712	-1.2335	-0.0605	7.0965	5.08%
1995	16,332	-0.8868	0.0240	7.2316	5.00%
1996	15,923	-0.9553	0.0207	7.6327	4.97%
1997	18,799	-0.8085	0.0155	7.7720	5.36%
1998	18,164	-0.6502	-0.0664	7.6026	5.06%
1999	19,543	-0.3911	-0.0794	7.2368	5.22%
2000	22,175	-0.0682	-0.1024	6.7181	5.12%
2001	22,695	0.0757	-0.1184	5.4978	5.25%
2002	22,461	0.2464	-0.0957	5.3638	4.85%
2003	22,831	0.4267	-0.0539	5.1639	4.87%
2004	21,790	0.7716	-0.0321	5.3452	4.92%
2005	21,793	0.7341	-0.0008	5.2593	5.14%
2006	20,534	0.9402	0.0381	5.5693	4.91%
2007	22,148	1.1377	0.0755	5.7011	5.27%
2008	23,454	0.7233	0.1670	5.0370	5.40%
2009	17,971	1.1788	0.1283	5.1727	5.25%
2010	18,599	1.1069	0.1798	7.3907	5.01%
2011	19,134	0.9084	0.1868	8.3084	4.98%
2012	21,603	0.8664	0.3699	7.4903	5.04%
2013	16,415	0.7798	0.4202	7.2127	4.90%

Period	Patents	MeanBreadth	MeanOrgExperience	MeanImpact	Top5Share
2014	9,657	0.7923	0.5188	7.0003	4.90%
2015	3,791	0.5775	0.8984	5.9330	5.22%
2016	539	-0.4169	1.3178	3.7625	4.82%

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	229,109	-2.5926	4.9416
Q2	76,899	-1.0495	5.3861
Q3	88,598	0.3530	5.8387
Q4	94,587	6.7691	10.379

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	z_value	p_value	model	vcov
breadth_c	0.0030	0.0158	0.1921	0.8477	poisson_baseline	cluster_gvkey_app_year
org_recomb_exp_log_c	0.0706	0.0354	1.9940	0.0461	poisson_baseline	cluster_gvkey_app_year
org_focus_c	-0.3609	0.3761	-0.9595	0.3373	poisson_baseline	cluster_gvkey_app_year
n_prior_art_cites	0.0018	0.0008	2.2149	0.0268	poisson_baseline	cluster_gvkey_app_year
recomb_degree	-0.4546	0.1934	-2.3506	0.0187	poisson_baseline	cluster_gvkey_app_year
field_recomb_exp_log	-0.1139	0.0457	-2.4923	0.0127	poisson_baseline	cluster_gvkey_app_year
total_patents_same_year	-0.0001	0.0001	-1.4188	0.1560	poisson_baseline	cluster_gvkey_app_year
total_patents_until_prev_log	-0.1661	0.0839	-1.9802	0.0477	poisson_baseline	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c	-0.0022	0.0008	-2.6692	0.0076	poisson_baseline	cluster_gvkey_app_year
breadth_c:org_focus_c	-0.0050	0.1059	-0.0474	0.9622	poisson_baseline	cluster_gvkey_app_year
org_recomb_exp_log_c:org_focus_c	0.3219	0.1779	1.8097	0.0703	poisson_baseline	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c:org_focus_c	-0.1705	0.0702	-2.4276	0.0152	poisson_baseline	cluster_gvkey_app_year
breadth_c	0.0147	0.0147	1.0030	0.3159	logit_top5	cluster_gvkey_app_year
org_recomb_exp_log_c	0.0988	0.0442	2.2329	0.0256	logit_top5	cluster_gvkey_app_year
org_focus_c	-0.8737	1.3579	-0.6434	0.5200	logit_top5	cluster_gvkey_app_year
n_prior_art_cites	0.0036	0.0019	1.9145	0.0556	logit_top5	cluster_gvkey_app_year

term	estimate	std_error	z_value	p_value	model	vcov
recomb_degree	-0.3229	0.1351	-2.3907	0.0168	logit_top5	cluster_gvkey_app_year
field_recomb_exp_log	-0.1319	0.0343	-3.8398	¡0.001	logit_top5	cluster_gvkey_app_year
total_patents_same_year	-0.0001	0.0000	-4.7631	¡0.001	logit_top5	cluster_gvkey_app_year
total_patents_until_prev_log	-0.3860	0.1008	-3.8305	¡0.001	logit_top5	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c	-0.0118	0.0023	-5.0773	¡0.001	logit_top5	cluster_gvkey_app_year
breadth_c:org_focus_c	-0.1306	0.3120	-0.4187	0.6754	logit_top5	cluster_gvkey_app_year
org_recomb_exp_log_c:org_focus_c	0.0361	0.8998	0.0401	0.9680	logit_top5	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c:org_focus_c	-0.2765	0.2994	-0.9236	0.3557	logit_top5	cluster_gvkey_app_year
breadth_c	-0.0099	0.0231	-0.4280	0.6686	logit_top1	cluster_gvkey_app_year
org_recomb_exp_log_c	0.1380	0.0934	1.4777	0.1395	logit_top1	cluster_gvkey_app_year
org_focus_c	-0.9087	3.6923	-0.2461	0.8056	logit_top1	cluster_gvkey_app_year
n_prior_art_cites	0.0048	0.0022	2.2062	0.0274	logit_top1	cluster_gvkey_app_year
recomb_degree	-0.4480	0.2543	-1.7618	0.0781	logit_top1	cluster_gvkey_app_year
field_recomb_exp_log	-0.2116	0.0677	-3.1246	0.0018	logit_top1	cluster_gvkey_app_year
total_patents_same_year	-0.0002	0.0001	-2.6951	0.0070	logit_top1	cluster_gvkey_app_year
total_patents_until_prev_log	-0.5362	0.2074	-2.5852	0.0097	logit_top1	cluster_gvkey_app_year
breadth_c:org_recomb_exp_log_c	-0.0132	0.0056	-2.3569	0.0184	logit_top1	cluster_gvkey_app_year
breadth_c:org_focus_c	-0.1517	1.2533	-0.1211	0.9036	logit_top1	cluster_gvkey_app_year
org_recomb_exp_log_c:org_focus_c	-0.5147	3.3074	-0.1556	0.8763	logit_top1	cluster_gvkey_app_year

term	estimate	std_error	z_value	p_value	model	vcov
breadth_c:org_recomb_exp_log_c:org_focus_c	-0.6392	1.0331	-0.6187	0.5361	logit_top1	cluster_gvkey_app_year

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Inventive impact	0	0.00%
Breadth	0	0.00%
Organizational experience	0	0.00%
Organizational focus	0	0.00%
Prior-art citations	0	0.00%
Recombination degree	0	0.00%
Field experience	0	0.00%
Same-year patents	0	0.00%
Firm age	0	0.00%

Missingness audit for the variables summarized in the manuscript.

Entity	Label	Observations	FirstPeriod	LastPeriod
6066	INTERNATIONAL BUSINESS MACHS COR	83,091	1986	2016
5047	GENERAL ELECTRIC CO	28,950	1986	2016
12141	MICROSOFT CORP	26,529	1994	2016
5606	HEWLETT PACKARD CO	24,186	1986	2016
7585	MOTOROLA INC	15,304	1986	2016
1300	ALLIED SIGNAL INC	14,534	1986	2016
10499	TEXAS INSTRUMENTS INC	14,212	1986	2016
4194	EASTMAN KODAK CO	12,898	1986	2010

Entity	Label	Observations	FirstPeriod	LastPeriod
7435	MINNESOTA MINING & MFG CO	12,696	1986	2016
4087	DU PONT E I DE NEMOURS & CO	11,004	1986	2016
4839	FORD MOTOR CO DEL	11,003	1986	2016
1690	APPLE COMPUTER INC	10,635	1986	2016
6266	JOHNSON & JOHNSON	10,285	1986	2016
2285	BOEING CO	10,165	1986	2016
10983	UNITED TECHNOLOGIES CORP	9,824	1986	2016
8762	PROCTER & GAMBLE CO	8,712	1986	2016
4060	DOW CHEMICAL CO	7,294	1986	2016
1704	APPLIED MATERIALS INC	6,975	1995	2016
5073	GENERAL MOTORS CORP	6,710	1986	2008
4503	EXXON CORP	6,538	1986	2016

Top entities by local observation count.

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